

Assignment - Day 5 - Register's and counters

- 1) Design a counter with T-flip flops whose count sequence is 0, 2, 4, 5, 3, 1, 0...

Present state (Q_2, Q_1, Q_0)	Next state (Q_2^+, Q_1^+, Q_0^+)	T_2	T_1	T_0
(0) 0 0 0	0 1 0	0	1	0
(2) 0 1 0	1 0 0	1	0	0
(4) 1 0 0	1 0 1	0	0	1
(5) 1 0 1	0 1 1	1	0	0
(3) 0 1 1	0 0 1	0	1	0
(1) 0 0 1	0 0 0	0	0	1

$Q_2 \backslash Q_1, Q_0$	00	01	11	10
$\bar{Q}_2$ 0	1	0	1	3
Q_2 1	4	5	X	6

$Q_2 \backslash Q_1, Q_0$	00	01	11	10
$\bar{Q}_2$ 0	1	0	1	3
Q_2 1	4	5	X	6

$$T_2 = Q_2 Q_0 + Q_1 \bar{Q}_0$$

$$T_1 = \bar{Q}_2 \bar{Q}_0 + Q_2 Q_0 + Q_1 \bar{Q}_0 + Q_1 Q_0$$

$Q_2 \backslash Q_1, Q_0$	00	01	11	10
$\bar{Q}_2$ 0	1	0	1	3
Q_2 1	4	5	X	6

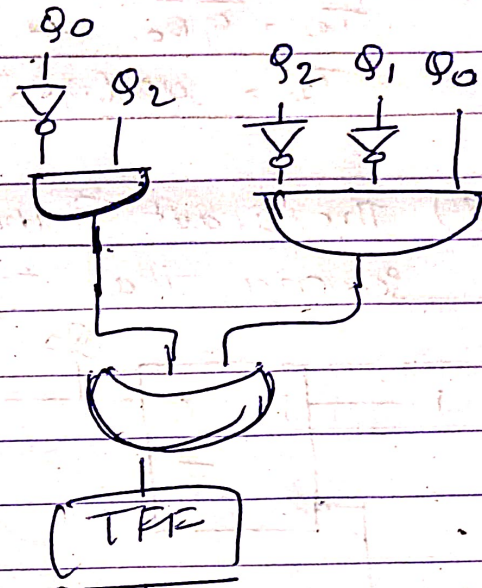
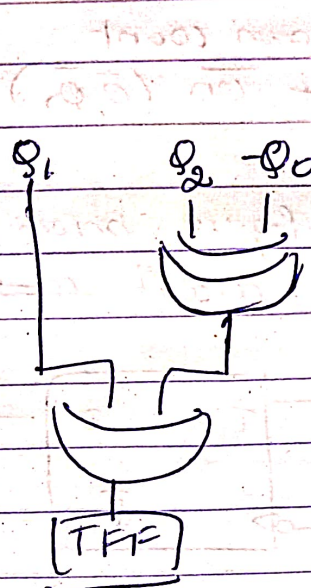
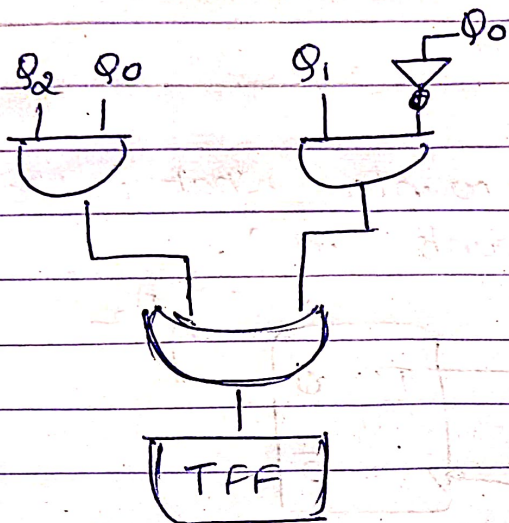
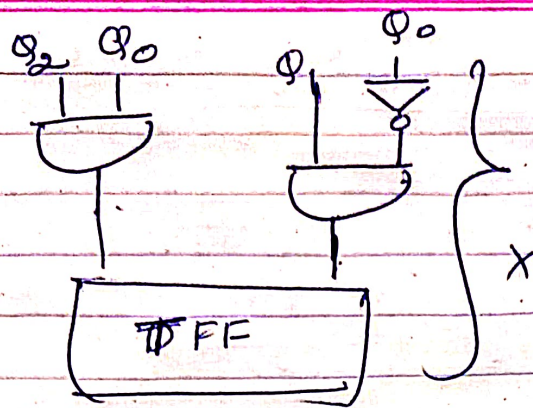
$$T_1 = \bar{Q}_2 \bar{Q}_0 + Q_1 + Q_2 Q_0$$

$$= Q_1 + (Q_2 \oplus Q_0)$$

$Q_2 \backslash Q_1, Q_0$	00	01	11	10
$\bar{Q}_2$ 0	0	1	3	2
Q_2 1	4	5	X	6

$$T_0 = Q_2 \bar{Q}_0 + \bar{Q}_2 Q_1 \bar{Q}_0 + Q_2 Q_1 Q_0$$

$$= \bar{Q}_0 Q_2 + \bar{Q}_2 Q_1 \bar{Q}_0$$



clk

2) Design a 3 bit updown counter using negative edge Triggered FF

Up count		Down count	
Present state	Next state	Present state	Next state
000	001	000	111
001	010	001	000
010	011	010	001
011	100	011	010
100	101	100	011
101	110	101	100
110	1101	110	101
111	000	111	110

$T_0 = 1$
 $T_1 = \phi_0$ } up count

$T_1 = \bar{\phi}_0 \rightarrow$ down counting

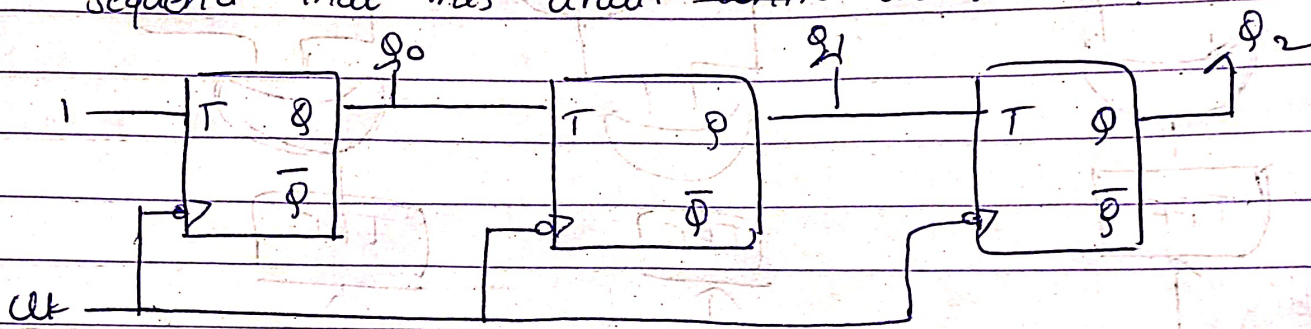
$$T_1 = \phi_0 \cdot \phi_0 + \bar{\phi}_0 \cdot \bar{\phi}_0$$

$T_2 = \phi_1 \phi_0 \rightarrow$ up count

$T_2 = \bar{\phi}_1 \bar{\phi}_0 \rightarrow$ down count

$$T_2 = \phi_0 (\phi_1 \phi_0) + \bar{\phi}_0 (\bar{\phi}_1 \bar{\phi}_0)$$

3.) The circuit in the figure below is a counter. What is the sequence that this circuit ~~count~~ counts in?



$T_0 = 1$, $T_1 = \phi_0$, $T_2 = \phi_1$

• Synchronous counter

$$\phi_0^+ = \phi_0 \oplus 1 = \bar{\phi}_0$$

$$\phi_1^+ = \phi_1 \oplus \phi_0$$

$$\phi_2^+ = \phi_2 \oplus \phi_1$$

Present state Next state

000

001

001

010

010

111

111

000

0 $\rightarrow$ 1 $\rightarrow$ 2 $\rightarrow$ 7 $\rightarrow$ 0